INCH-POUND

MIL-S-48687 (AR)
AMENDMENT 1
16 February 1990

MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143E1
LOADING, ASSEMBLING AND PACKING FOR

This amendment forms a part of Military Specification MIL-S-48687 (AR), dated 4 February 1987, and is approved for use within the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 7

Table IV, g & h: Delete "2.6" (six places) and substitute "4.0" (six places).

PAGE 8

3.7.1, Line 7: Delete "2.5" and substitute "1.5".

3.7.2, last line: Delete "2.5" and substitute "4.0".

3.7.4: Delete in its entirety and substitute:

"3.7.4 Fire circuit function. The initiation of either the left or right fire circuits functioning is determined by the occurrence of the fire TM output changing from 0 ± .5 VDC to 4.5 ± 1.0 VDC at (+) J1-20 to (-) J1-15. The left fire circuit has completed functioning when the left fire circuit capacitor voltage (+) J1-38 to (-) J1-15 has discharge to less than 15.0 VDC in no greater than 400 milliseconds after the occurrence of a fire TM output. The right fire circuit has completed functioning when the right fire circuit capacitor voltage (+) J1-36 to (-) J1-15 has discharge to less than 15.0 VDC in no greater than 400 milliseconds after the occurrence of a fire TM output."

PAGE 9

3.7.4.1.1: Delete in its entirety and substitute:

"3.7.4.1.1 Loss of power after arming. The initiation of fire circuit functioning as defined by paragraph 3.7.4 shall occur in no less than 10 milliseconds and no greater than 50 milliseconds upon removal of power/ destruct (+) J1-30 to (-) J1-15."

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3.7.4.1.2: Delete the last sentence and substitute:

"The fire circuits shall function as described in paragraph 3.7.4. Both the left and right fire circuit capacitor voltages shall have discharge to less than 15.Ø VDC in 1.2 seconds maximum after the initial fire TM voltage transitioned as described above."

3.7.4.2.4, line 4: Delete "then".

PAGE 12

3.9.1, Table VI: Under minimum and maximum

(OHMS)	delete		and substitute	
	6.1	8.2	5.9	8.1
	4.8	6.5	4.6	6.2
	7.2	9.2"	7.3	9.1"

PAGE 19

3.12.4: Delete in its entirety and substitute:

"3.12.4 Lateral arming. When a Safe and Unlocked S&A Device with 5.Ø + 1.Ø VDC applied to 5V Test (+) J1-1 to (-) J1-15, is subjected to the acceleration environment specified below, both detonator rotors shall arm in 2.8-4.Ø seconds and remain Armed for the duration of the application of the acceleration. Arming is defined as the detonator rotors rotating from the Safe and Unlocked position to the Armed position when Rotor Monitor (Right) (+) J1-18 to (-) J1-15 and Rotor Monitor (Left) (+) J1-2 to (-) J1-15 have changed from 4.5 ± 1.Ø VDC to Ø.Ø ± Ø.5 VDC. A minimum of one Detonator Rotor shall be Armed and Locked after the removal of the acceleration. The side required to be locked shall be the (left) side when the direction of the applied lateral acceleration is in the (+Y) axis. The side required to be locked shall be the (right) side when the direction of the applied lateral acceleration is in the (-Y) axis (refer to Figure 1).

a. An acceleration profile along the first transverse axis (+Y, -Y) of the S&A which is equal to the lateral acceleration profile in drawing 9357Ø7.

b. An acceleration along the longitudinal axis of the S&A in the (-Z) direction, which is equal to the axial acceleration profile in drawing 9357Ø7.

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c. Acceleration components to be applied simultaneously to the S&A. Maximum permissible variation in centrifuge revolutions per minute (RPM) is ± 8%. Maximum permissible variation in angular orientation of the S&A is ± 4% (See 4.5.11.1.1)."

3.12.5: Delete "a and b" and substitute:

"Vector sum of:

a. 46.8g longitudinal aft-to-forward (-Z) and 52.0g lateral (+Y or -Y) (=70.0g at 48° from longitudinal axis).

b. 22.3g longitudinal forward-to-aft (+Z) and 52.5g lateral (+Y or -Y) (=57.0g at 67° from longitudinal axis)."

PAGE 21

3.14.11, last line: Delete "2.5" and substitute "1.5".

PAGE 22

3.14.14: Delete in its entirety.

PAGE 25

4.3.1, a, Under part description: Delete "Electrical Assembly" and substitute "Detonator Electrical Assembly".

4.3.1, m: Delete in its entirety and substitute: "EM assembly, 12630801, 10".

4.3.1. o, Under part description: Delete "M143" and substitute "M143E1".

PAGE 27

4.3.4, Under paragraph reference/inspection method: Delete "4.4.2.31" and substitute "4.4.2.33".

PAGE 29

4.4.2.1, Sheet 1 of 1, next higher assembly: Delete "12630800" and substitute "12630858".

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PAGE 30

4.4.2.2, Sheet 1 of 1, Major 102, under Examination or Test: Add "(2 places)".

PAGE 38

4.4.2.10, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 42

4.4.2.14, Sheet 1 of 1, Major 102: Delete in its entirety.

PAGE 43

4.4.2.15, Sheet 1 of 1: Delete in its entirety.

PAGE 44

4.4.2.16, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 46

4.4.2.18, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 48

4.4.2.20, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 49

4.4.2.21, Sheet 1 of 1, Major 105, under requirement paragaph: Delete "3.9.1" and substitute "3.9.1a".

PAGE 53

4.4.2.25, Sheet 1 of 2, Major 106, under requirement paragraph: Delete "3.9.3".

PAGE 54

4.4.2.25, Sheet 2 of 2, Under Examination or Test: Delete "12630800" and substitute "12630865-3".

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48687 (AR)
AMENDMENT 1

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311077
4.4.2.10	Lead Assembly				NEXT HIGHER ASSEMBLY 12630861
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Evidence of cracks at crimp		100%	3.2	Visual
Major					
101	Length		0.40%	3.2	Gage
102	Diameter, max.		0.40%	3.2	Gage
103	Radiographic Inspection		0.40%	3.5.1	MIL-STD-202, M209
104	Output		0.40%	3.5.2	4.5.10.2/Test
Minor					
201	Missing or excessive green laxquer		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48687 (AR)
AMENDMENT 1

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630868
4.4.2.16	Gasket				NEXT HIGHER ASSEMBLY 12630801
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter of hole (4 places)		Ø.65%	3	Gage
202	Location of hole (4 places)		Ø.65%	3	Use P/N 12630849
203	Dimensions of cut-out		Ø.65%	3	as a functional
204	Location of cut-out		Ø.65%	3	gage 1/
205	Evidence of poor workmanship		1.0%	3.18	Visual

NOTES:
1/ 50% or more of each of the 4 mounting holes shall be visible through the gasket.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48687 (AR)
AMENDMENT 1

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630800
4.4.2.18	Safety & Arming Device: M143E1 (Loaded) After Installing Lead Assembly				NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lead assembly mounting screws missing or loose		0.40%	3.2	Visual/Manual
102	Logic board mounting screws missing or loose (4 places)		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		1.0%	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48687 (AR)
AMENDMENT 1

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.20	Electro-Mechanical Assembly (Visual Prior to Logic & Fire Board Assembly)		1 1		12630801
					NEXT HIGHER ASSEMBLY 12630864/12630800
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Shunt wire not removed from resistor shorting plate detonator circuit (2 places)		100%	3.6.1	Visual
102	Resistance (2 places)		100%	3.6.1	Test

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.2.27, Sheet 1 of 1, Major 105 and 106: Delete in their entirety.

PAGE 57

4.4.2.28, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 60a

Add: "4.4.2.32, Sheet 1 of 1 with included sheet".

PAGE 60b

Add: "4.4.2.33, Sheet 1 of 1 with included sheet".

PAGE 62

4.5.2, title: Delete in its entirety and substitute the following:

"Procedure for Function on Loss of Power after Arming Test".

4.5.2.1: Add "e. 5.0 + 1.0 VDC Applied to 5V Test input".

4.5.2.2: Delete "a. and b." and substitute:

"a. Perform arming as specified in 4.5.7.2.

b. Remove 31.0 + 0.5 VDC on Power/Destruct.

c. Measure Time (3.7.4.1.1) to initiation of Fire Circuit Functioning.

d. Measure Time (3.7.4) for L & R Fire Circuits to Function".

4.5.3.1: Add "f. 5.0 + 1 OVDC Applied to 5V Test input".

4.5.3.2, Under b.: Delete "4.5.7" and substitute "4.5.7.2".

4.5.3.2, Under c.: Delete in its entirety and substitute:

"c. Measure Time (3.7.4.1.2) to change of rotor monitor with respect to fire TM and verify capacitor voltage left and right is less than 15.0 VDC at 1.2 seconds after initial occurrence of fire TM."

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	Safety & Arming Device: M143E1 (Loaded) Electrical Operation, CRAF Test Set	1 OF 1	12630800
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED				
Critical					
1	Safety (Safe & Locked)	100%		3.11.1.2.1	Test/4.5.1.2
Major					
101	Continuity	100%		3.14.1	Test
102	Resistance	100%		3.14.2	Test/3.9.1a
103	Connector Pins to Case Ground Isolation Resistance	100%	1/	3.14.3	Test
104	Fire Circuit Charging (charge command)	100%		3.14.11	Test
105	Fire Circuit Capacitor Voltage Storage	100%		3.14.12	Test
106	Voltage Outputs (Safe & Locked)	100%		3.14.13	Test
107	Fire Circuit Discharging	100%		3.14.14	Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.32	LAT Shorting Plate Detonator Housing Assembly		1 1	939905-01
				NEXT HIGHER ASSEMBLY: 12630865

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Detonator not flush after assembly	1/	100%	3.2	Visual
202	Detonator leads not twisted together		100%	3.2	Visual
203	Insulation sleeving missing		100%	3.2	Visual
204	100 ohm resistor missing or incorrect value		100%	3.2	Visual
205	Evidence of solder flow into contact area		100%	3.2	Visual
206	Evidence of poor workmanship		100%	3.18	Visual

NOTES: 1/ Two times the number of LAT samples.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48687 (AR)
AMENDMENT 1

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.33	LAT Shorting Plate Detonator Housing Assembly	1	1	939905-01
				NEXT HIGHER ASSEMBLY: 12630865

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Detonator not connected	1/	100%	3.2	Visual
202	Insulation sleeving missing		100%	3.2	Visual
203	Solder not in accordance with MIL-STD-1460		100%	3.2	Visual
204	Evidence of solder flow into contact area		100%	3.2	Visual
205	Evidence of poor workmanship		100%	3.2	Visual

NOTES: 1/ Two times the number of LAT samples.

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.5.3.2: Add "d. Measure Time (3.7.4) for left and right Fire Circuits to Function."

PAGE 64

4.5.8.2, line 7: Delete "both detonators" and substitute "both detonator rotors".

PAGE 65

4.5.10: Delete "." and substitute "/S&A (loaded)".

4.5.10.1, line 1, after Assembly: Add "or S&A (loaded)".

4.5.10.1, CAUTION, line 1, after Assembly: Add "or S&A (loaded)".

4.5.10.2, line 1, after assembly: Add "or S&A (loaded".

4.5.10.2, CAUTION, line 1, after Assembly: Add "or S&A (loaded)".

PAGE 70

4.5.11.1.1: Add "The recommended method for demonstrating compliance to these requirements is sampling both R.P.M. and angular orientation in intervals no greater than 50 milliseconds throughout the duration imparted to the S&A device."

PAGE 71

6.2.2, last line: Delete "Dover, New Jersey 07801-5001" and substitute "Picatinny Arsenal, New Jersey 07806-5000".

PAGE 72

6.6: Delete "$\sqrt{\frac{9\times10^{9}}{R\times28416}}$" and substitute "$187.6\sqrt{\frac{g}{R}}$".

PAGE 74

APPENDIX A, n, under drawing: Delete "12630847" and substitute "9311370".

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1336-A575)

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